

Computer & Programming Fundamentals

Unit-1

Computer

A computer can be defined as an electronic device that accepts raw data as input and processes it with a set of instructions (a program) to produce the result as output. It renders output after performing mathematical and logical operations and can save the output for future use.

Generations of Computer

Generation refers to development that changes the way computers work.

Generations of computer are mainly divided according to the development of computer technology.

First Generation Computers: Vacuum Tubes (1940-1956)

- The technology behind the primary generation computers was a fragile glass device, which was called **vacuum tubes**.
- First-generation computers were used for calculation, storage, and control purpose.
- They needed a full room and consume lot of electricity.

Main first generation computers are:

ENIAC:

Electronic Numerical Integrator and Computer. It was very heavy, large and contained 18,000 vacuum tubes.

EDVAC:

Electronic Discrete Variable Automatic Computer was designed by von Neumann.

UNIVAC:

Universal Automatic Computer

Advantages of the first-generation computer

1. These computers were in no time in terms of calculating. They might calculate in milliseconds.
2. Vacuum tubes are the electronic components available at that time, which were used by these computers.

Disadvantages of the first-generation computer

1. These computers' weight was about 30 tones and took up a lot of space as they were very big in size.
2. These computers were very costly and were non portable.
3. Due to the presence of magnetic drums, they were only able to store a small amount of information.
4. As the first generation of computers were used vacuum tubes, which need a large cooling system.

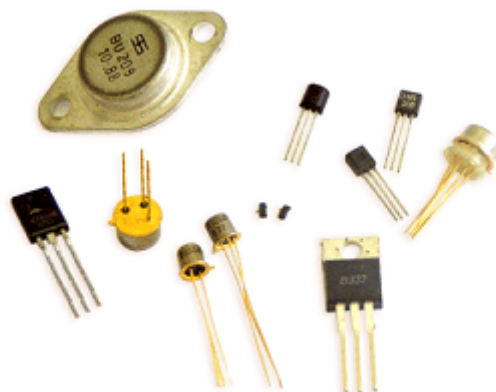
5. They consumed a large amount of energy and had less work efficiency.
6. They needed punch cards to give input.



Second Generation Computers: Transistors (1956-1963)

- Instead of vacuum tubes, the second generation of computers was supported transistors.
- From 1956 through 1963, transistors were widely utilized within the second generation of computers.
- These computers were small in size because they used transistors in these generations of computers that were smaller than vacuum tubes.
- In terms of speed, the second generation of computers was faster.
- They were cheaper to create.

- Some samples of the second generation of computers are given below:
 - CDC 3600
 - Honeywell 400
 - UNIVAC 1108
 - IBM 7094
 - CDC 1604, and many more



Transistors

Advantages of the second generation of computers

1. Small in size as compared to the previous generation of computers.
2. These computers consumed less energy and did not produce the maximum amount of heat.
3. In these computers, punch cards and Assembly language were used to give input.
4. As compared to the first generation, it had better portability and low cost.
5. Furthermore, these computers were faster in speed as could calculate data in microseconds.

Disadvantages of the second generation of computers

1. It required a cooling system and constant maintenance.
2. Also, these computers were used only for a particular objective.

Third Generation Computers: Integrated Circuits(1964-1971)

- Technology shift from huge transistors to integrated circuits, also referred to as IC.
- The most important feature of this era's computer was the speed and reliability.
- IC was made from silicon and also called silicon chips.
- Programming was now wiped out Higher level languages like BASIC (Beginners All-purpose Symbolic Instruction Code).

Some examples of third generation computers are:

- IBM 370
- PDP-8
- ICL 2900
- IBM 360
- PDP-11, and many more



Advantages of the third generation of computers

1. These computers were cheaper in cost.
2. This generation's computers used Integrated circuits that made them small in size. Also, they were reliable and faster in speed.
3. The use of an Integrated circuit was not only beneficial for reducing the size of the computer; it also enhances the performance of the computer.
4. The third generation of computers introduced a big storage capacity.
5. In these computers, keyboard and mouse were used to give the input rather than punch cards.
6. These generation computers were much fast in terms of calculation; they decrease computational time from microseconds to nanoseconds.
7. For better performance, these computers used operating systems and used multiple programming and the concept of time-sharing.

Disadvantages of the third generation of computers

1. Air conditioning was required by these computers.
2. The manufacturing of IC chips required a highly sophisticated technology.
3. IC chips are not easy to maintain.

Fourth Generation Computers: Micro-processors (1971-2010)

- The large scale of integration LSI circuits built on one chip called microprocessors.

The Microprocessor, commonly called a CPU (Central Processing Unit), was used by the fourth generation of computers.

- The most advantage of this technology is that one microprocessor can contain all circuits required to perform arithmetic, logic and control functions on one chip.

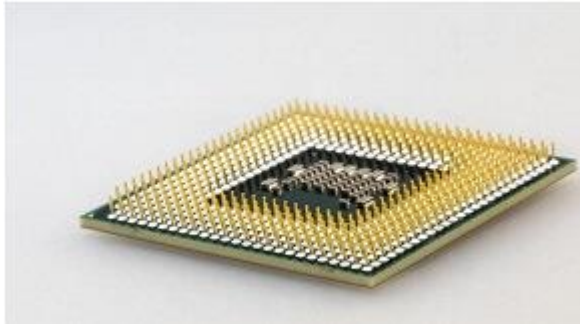
- Technologies like multiprocessing, multiprogramming, time-sharing, operating speed and virtual memory made it a more user-friendly and customary device.

Microprocessors, along with integrated circuits, helped to introduce the laptop.

Some examples of fourth-generation computers are:

- STAR 1000
- PUP 11
- IBM 4341

- DEC 10



Advantages of the fourth generation of computers

1. This generation computer size gets reduced and fastest in computation, comparing to the previous generation of computer.
2. Its heat generated is negligible and required less maintenance.
3. In these types of computers, all kinds of high-level language can be used.

Disadvantages of the fourth generation of computers

1. The fabrication and design of the microprocessor are not simple.
2. In many cases, air conditioning is needed due to the occurrence of Integrated circuits.
3. For making the Integrated circuit, advanced technology is required.

Fifth Generation Computers (2010 to present)

- It allows computers to behave like humans.

The fifth generation of computers is predicated on artificial intelligence.

- It is often seen in programs like voice recognition, area of medicines, and entertainment.
- Within the field of games playing also it's shown remarkable performance where computers are capable of beating human competitors.
- The speed is highest, size is that the smallest.

The ten million electronic components have consisted of the production of a microprocessor as this generation is based on Ultra Large-Scale Integration (ULSI).

Some examples are:

- UltraBook
- Desktop
- Laptop
- NoteBook

Advantages of the fifth generation of computers

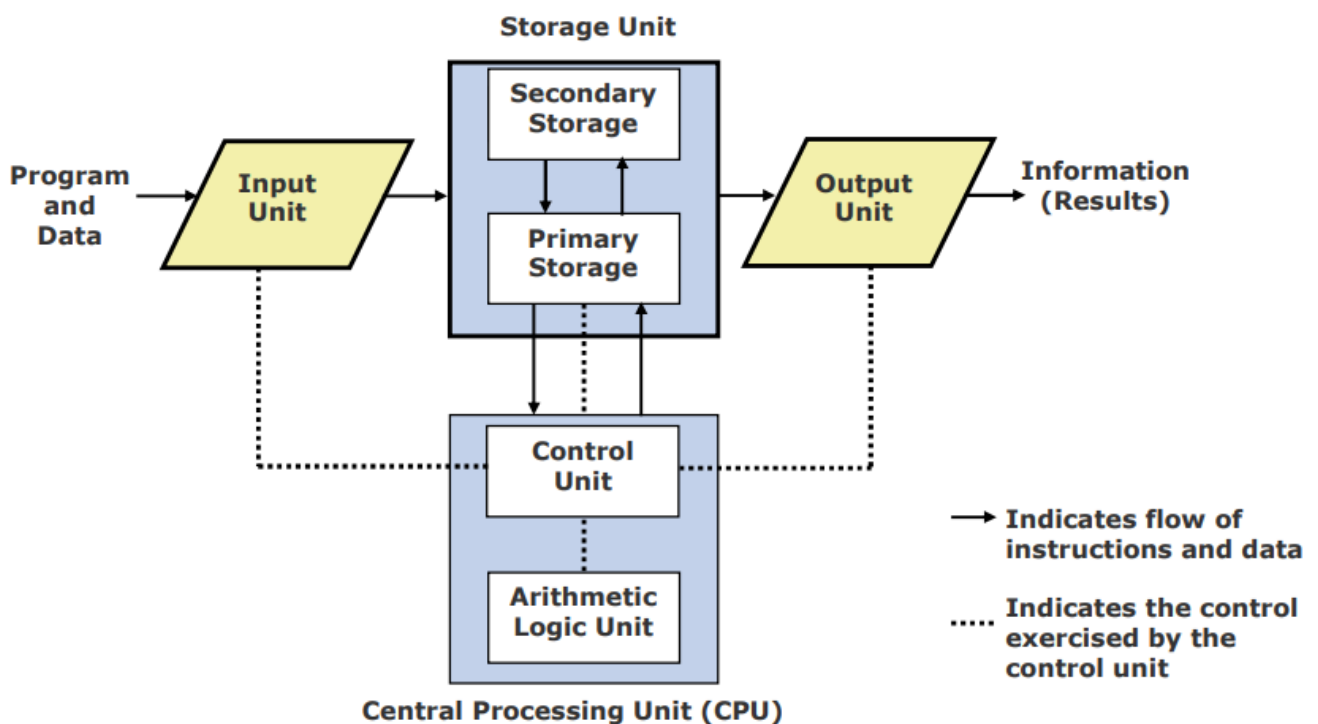
1. It has unique features and is available in several sizes.
2. It is more reliable and has the ability to work much faster.

3. It offers computers that have a more friendly UI (User Interface), including multimedia features.

Disadvantages of the fifth generation of computers

1. In these computers, very low-level languages are required.
2. They may be harmful in terms of making human brains doomed and dull.

Components Of Computer



Block Diagram Of Computer

There are 4 main computer components that are given below:

- Input Unit
- Output Unit
- Memory Unit
- CPU

Input Unit

1. This unit is responsible for handling all the inputs to the computer.
2. It accepts (or reads) instructions and data from outside world.
3. It converts these instructions and data in computer acceptable form
4. It supplies the converted instructions and data to the computer system for further processing.
5. Input devices include keyboard, mouse, joystick etc.

Output Unit

1. This unit is responsible for representing the output to the user of computer.
2. It converts these coded results to human acceptable (readable) form
3. It supplies the converted results to outside world

4. Output devices include monitor, printer, speakers etc.

Memory Unit

1. The data is first stored in storage unit for faster access and processing.
2. It stores data and instructions required for processing (received from input devices)
3. Intermediate results of processing
4. Final results of processing, before they are released to an output device

Primary Storage

- Primary storage is used to hold running program instructions
- Used to hold data, intermediate results, and results of ongoing processing of job(s)
- Fast in operation
- Small Capacity
- Expensive
- Volatile (loses data on power dissipation)
- Example- RAM and ROM(non volatile)

Secondary Storage

- Used to hold stored program instructions
- Used to hold data and information of stored jobs
- Slower than primary storage

- Large Capacity and lot cheaper
- Non volatile (Retains data even without power)
- Example- CD, hard drive etc

CPU – Central Processing Unit

- It is the brain of a computer system.
- It is responsible for controlling the operations of all other units of a computer system.
- It executes the calculations and commands required for functioning of the computer device.
- The CPU comprises some components: the control unit, the arithmetic logic unit (ALU).

Arithmetic Logic Unit of a computer system is the place where the actual executions of instructions takes place during processing operation.

The ALU operates addition, subtraction, comparison calculations, and other logical and mathematical processes.

Control Unit of a computer system manages and coordinates the operations of all other components of the computer system

Characteristics of Computer System

1. Speed

The processing speed of computer is very high. Executing mathematical calculation, computer works faster and more accurately than human. A computer is a time saving device.

2. Diligence

A human cannot work for several hours without resting, yet a computer never tires. A computer can conduct millions of calculations per second with complete precision without stopping. A computer can consistently and accurately do millions of jobs or calculations.

3. Reliability

A computer is reliable. The output results never differ unless the input varies. the output is totally depend on the input. When an input is the same the output will also be the same.

4. Automation

The world is quickly moving toward AI (Artificial Intelligence)-based technology. A computer may conduct tasks automatically after instructions are programmed. By executing jobs automatically, this computer feature replaces thousands of workers.

5. Versatility

Versatility refers to a capacity of computer. Computer perform different types of tasks with the same accuracy and efficiency. A computer can perform multiple tasks at the same time this is known as versatility. For example, while listening to music, we may develop our project using PowerPoint

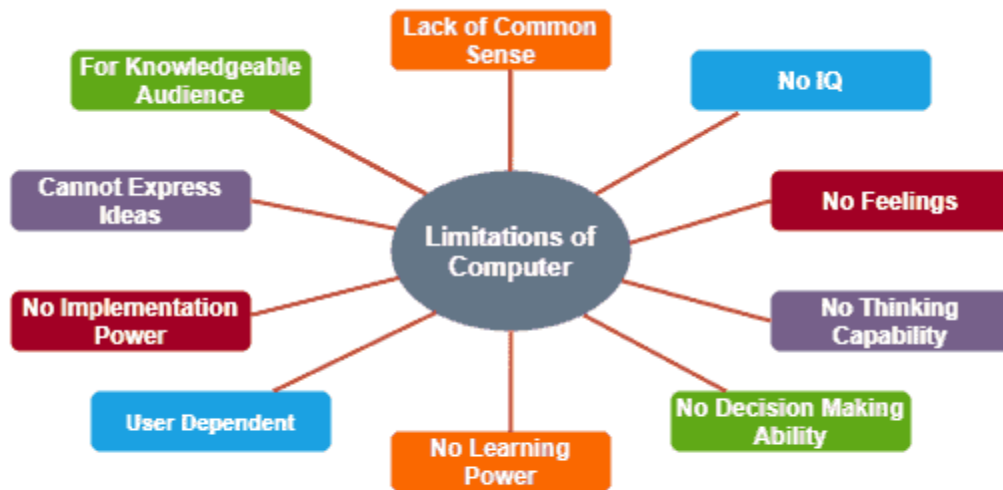
6. Memory

A computer can store millions of records. These records may be accessed with complete precision. Computer memory storage capacity is measured in Bytes, Kilobytes(KB), Megabytes(MB), Gigabytes(GB), and Terabytes(TB). A computer has built-in memory known as primary memory.

7. Accuracy

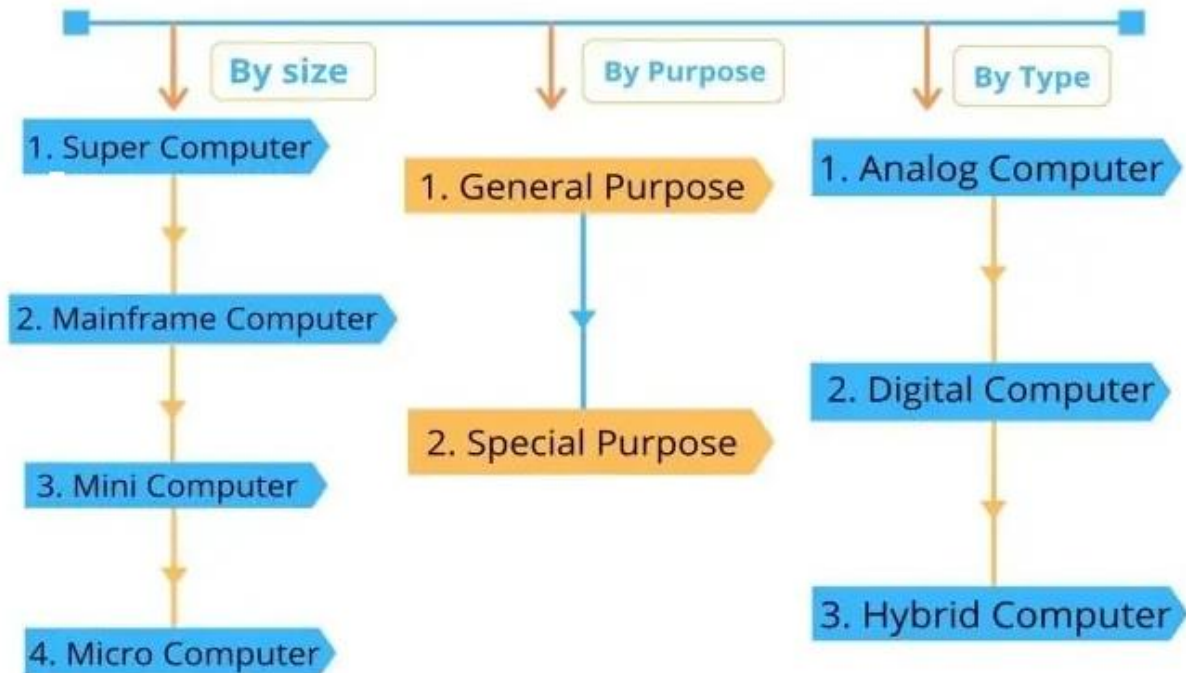
When a computer performs a computation or operation, the chances of errors occurring are low. Errors in a computer are caused by human's submitting incorrect data. A computer can do a variety of operations and calculations fast and accurately.

Limitations Of Computer



Limitations of Computer

classification of computers



1) Supercomputer

- i. Supercomputers are the ***biggest and fastest computers***.
- ii. They are designed to process huge amount of data. A supercomputer can ***process trillions of instructions in a second***.
- iii. Supercomputers are particularly used in ***scientific and engineering applications*** such as weather forecasting, scientific simulations and nuclear energy research.

Characteristics of SuperComputers:

- It produces excellent results in animations.
- It is used for virtual testing of nuclear weapons and critical medical tests.
- It can study and understand climate patterns and forecast weather conditions.
- It helps in designing the flight simulators for pilots at the beginner level for their training.
- It has played a vital role in managing the online currency world such as stock market and bitcoin.
- It helps in the diagnosis of various critical diseases and in producing accurate results in brain injuries, strokes, etc.
- It helps in scientific research areas by accurately analysing data obtained from exploring the solar system, satellites, and movement of Earth.

- It also used in a smog control system where it predicts the level of fog and other pollutants in the atmosphere.

2.Mainframe computers

Mainframe computers are designed to ***support hundreds or thousands of users simultaneously***. They can ***support multiple programs*** at the same time. These features of mainframe computers make them ideal for big organizations like banking and telecom sectors, which need to manage and process a high volume of data.

Characteristics of Mainframe Computers:

- It can process huge amount of data, e.g. millions of transactions in a second in the banking sector.
- It has a very long life. It can run smoothly for up to 50 years after proper installation.
- It gives excellent performance with large scale memory management.
- It has the ability to share or distribute its workload among other processors and input/output terminals.
- There are fewer chances of error or bugs during processing in mainframe computers. If any error occurs it can fix it quickly without affecting the performance.
- It has the ability to protect the stored data and other ongoing exchange of information and data.

Applications of mainframe computers:

- In ***health care***, it enabled hospitals to maintain a record of their millions of patients in order to contact them for treatment or related to their appointment, medicine updates or disease updates.
- In the ***field of defence***, it allows the defence departments to share a large amount of sensitive information with other branches of defence.
- In the ***field of education***, it helps big universities to store, manage and retrieve data related to their courses, admissions, students, teachers, employees and affiliated schools and colleges.
- In ***retail sector***, retail companies have a huge customer base and branches use mainframe computers to handle and execute information related to their inventory management, customer management and huge transactions in short duration.

3) Minicomputer

It is a ***midsize multiprocessing computer***. It consists of two or more processors and can support ***4 to 200 users at one time***. Mini computers are used in institutes and departments for tasks such as billing, accounting and inventory management. A minicomputer ***lies between***

the mainframe and microcomputer as it is smaller than mainframe but larger than a microcomputer.

Characteristics of minicomputer:

- It is light weight that makes it easy to carry and fit anywhere.
- It is less expensive than mainframe computers.
- It is very fast compared to its size.
- It remains charged for a long time.
- It does not require a controlled operational environment.

Applications of minicomputers:

- **Process control:** It was used for process control in manufacturing. It mainly performs two primary functions that are collecting data and feedback. If any abnormality occurs in the process, it is detected by the minicomputer and necessary adjustments are made accordingly.
- **Data management:** It is an excellent device for small organizations to collect, store and share data. Local hospitals and hotels can use it to maintain the records of their patients and customers respectively.
- **Communications Portal:** It can also play the role of a communication device in larger systems by serving as a portal between a human operator and a central processor or computer.

4) Microcomputer

Microcomputer is also known as a personal computer. It is a general-purpose computer that is designed for individual use. It has a microprocessor as a central processing unit, memory, storage area, input unit and output unit. Laptops and desktop computers are examples of microcomputers. They are suitable for personal work that may be making an assignment, watching a movie, or at office for office work.

Characteristics of a microcomputer:

- It is smallest in size among all types of computers.
- A limited number of software can be used.
- It is designed for personal work and applications. Only one user can work at a time.
- It is less expensive and easy to use.
- It does not require user to have special skills or training to use it.
- It is capable of multitasking such as printing, scanning, browsing, watching videos, etc.

1) Analogue Computer

- Analogue computers are designed to ***process analogue data***.
- Analogue data is continuous data that changes continuously and cannot have discrete values.

- Analogue computers are used where we don't need exact values always such as speed, temperature, pressure and current.
- Analogue computers directly accept the data from the measuring device without first converting it into numbers and codes.
- They measure the continuous changes in physical quantity and generally render output as a reading on a scale.
- **Speedometer** and **mercury thermometer** are examples of analogue computers.

2) Digital Computer

- Digital computer is designed to perform calculations and logical operations at high speed.
- It accepts the raw data as input in the form of digits or binary numbers (0 and 1) and processes it with programs stored in its memory to produce the output.
- All modern computers like laptops, desktops including smartphones that we use at home or office are digital computers.

Advantages of digital computers:

- It allows you to store a large amount of information and to retrieve it easily whenever you need it.

- You can easily add new features to digital systems more easily.
- Different applications can be used in digital systems just by changing the program without making any changes in hardware
- The cost of hardware is less due to the advancement in the IC technology.
- It offers high speed as the data is processed digitally.
- It is highly reliable as it uses error correction codes.
- Reproducibility of results is higher as the output is not affected by noise, temperature, humidity, and other properties of its components.

3) Hybrid Computer

- Hybrid computer has features of both analogue and digital computer.
- It is ***fast like an analogue*** computer and has memory and ***accuracy like digital computers***.
- It can process both continuous and discrete data.
- It accepts analogue signals and convert them into digital form before processing.
- It is widely used in specialized applications where both analogue and digital data is processed.
- For example, a processor is used in petrol pumps that converts the measurements of fuel flow into quantity and price.

Advantages of using hybrid computers:

- Its computing speed is very high due to the all-parallel configuration of the analogue subsystem.
- It produces precise and quick results that are more accurate and useful.
- It has the ability to solve and manage big equation in real-time.
- It helps in the on-line data processing.

1.General Purpose Computer

- A general-purpose computer system is one that can be programmed to perform a wide range of functions.
- Writing a word processing letter, document preparation, recording, financial analysis, printing papers, constructing databases, and computations are all activities that general computers can perform with precision and consistency.
- These computers are often smaller
- They have less storage space, and are less expensive.

2.Special Purpose Computer

- These computers are made to carry out a specific or specialised activity.
- The scope and scale of the job depends on the size, storage capacity, and cost of such computers.

- The purpose of these computers is to do a certain task.
- Weather prediction, planetary science, farming, engineering, meteorology, satellite operation, traffic control, and pharmaceutical science research all utilize these computers.



Uses Of Computer System

1.Business

Computers can be seen in almost every business. Computers are almost part of a business setup because they increase productivity. In businesses, computers are used to store and manage accounts and personal data, maintain projects, track status, and make reports and presentations. Computers also help people analyze their investment, expenses, profits, sales etc.

2.Science

In science, research and engineering, computers are best suited for collecting, analyzing, categorizing, and storing the data. Computers enable scientists from different locations to work together on the same project. Computers play role in launching, maintaining, controlling spacecraft, operating and other technologies.

3.Government

In government sector, they are used to perform various functions in different departments and improving their services' quality, efficiency, and productivity. Some examples of such services are city planning, traffic control, law enforcement, infrastructure developments, and tourism. The primary purposes of using computers are data processing tasks, maintaining citizens' database, and promoting a paperless environment. Computers are

playing crucial role in country's defense system. They help in missile development, rocket, satellite launches.

4.Health and Medical

Computers are changing the methods of diagnosis in hospitals. They are used for maintaining patients' information, records, live monitoring of patients, X-rays. Computers help monitor heart rate, and blood pressure, etc. Computers enable doctors to exchange patient's data easily with other medical specialists. Advanced surgical devices are based on robotics that helping surgeons to conduct operations and surgeries remotely.

5.Education

Computers are broadly getting used in the education field. They help people to get different educational materials such as images, videos, e-books in one place. All such information can be accessed through Internet. Computers are best suited for online classes, online tutoring, online examinations, and creating assignments and projects. They can also be used to maintain and monitor student performance and other information.

6.Industry

Computers are used in industries to perform various tasks such as maintaining inventory, interior designing, designing samples, communicating over video conferencing etc. Online marketing has made it easier for people to buy products in rural areas. Computers have

enabled people from different levels of different locations to participate easily in stock marketing.

7.Banking

Most countries use online banking systems where customers can access their data directly using computers and the Internet. People can check their account balance, transfer money, and pay online bills, including credit cards. Banks use computers to perform transactions and store customer data, transaction records, etc. Banks have reduced the number of manual errors, number of employees, and costs to a great extent by using computers. ATMs are example of computers that are helping people to withdraw and deposit the money.

8.Entertainment

Computers nowadays are one of the best mediums for entertainment. Computers can be used to watch movies, play games, listen to music, etc. People can also enjoy recording their videos with webcam and apply several entertaining AI effects. Several Photo editor programs are also available with fabulous powerful features.

9.Training

Most companies use computers to provide training to their employees. Computer-based training helps companies save their time, money and increase productivity. Computer-based training can be used to train employees for large distances in various locations.

This will eliminate travel time and costs, making training process more comfortable and smoother.

10.Arts

Computers have become part of art, photography, dance, and culture. Computers with advanced features allow users to draw their projects directly on it. Besides, people can use computers to digitize their photos. There are several photo editor software that can help people edit and customize their photos. The dance's movements and steps can be shown live with animations' help.

11.Sports

In today's developed world, computers are being used in almost every sport. In sports, computers are mainly used to maintain scoreboards, records, and other statistics. They are used to analyze player movements and make various in-game decisions. Computers help to make complex in-game decisions (especially in umpiring), which cannot be seen by human eyes.

12.Robotics

Robotics is one of the emerging fields of technology that uses computers for science and engineering and designing machines. These machines can be virtual and physical that can reduce or eliminate human workload. Additionally, some machines can perform heavy tasks that humans cannot complete, or that may take a long time to complete. Car manufacturing was one of the first

examples where robots helped to assemble car parts and perform many other heavy tasks. Robots are beneficial in many fields, such as exploring areas where conditions are difficult for humans, helping military, helping law enforcement and helping health professionals, etc.

13.Safety and Security

Computers are capable of working together with different equipment, tools and technologies. When it comes to safety, computers are widely used with security camera. Every private and government organizations are equipped with security cameras to monitor people and goods. These cameras are helping security intelligence agencies to identify terrorists or criminals in public places. Computers attached with a fingerprint scanner and face-recognition technology have made it harder for fraudsters to use fake identities and take government benefits using someone else's identity. Security systems have become so powerful with introduction of computer technology and the Internet.

14.Publishing

Computers are one of the first choices when it comes to publication. Computers are mainly used for designing newsletters, novels, posters, magazines, newspapers etc. They are suitable for almost every type of publication. Computers used in both types of publishing methods,

either its hard-copy or e-books. Computers play a vital role in blogging and writing articles on websites.

15.Weather Forecasting

It is almost impossible for humans to predict the weather without using a computer. The weather forecasting process involves complex computation and monitoring of data from satellites and many other technological devices. The computer has made it a little easier to predict weather conditions because it can process information from different sources and make complex calculations related to it, which are necessary for forecasting. Computers process enormous amounts of meteorological information during weather forecasting.

16.Communication

Computers have become primary sources of communication. Modern computers come with built-in webcam and microphone to make use of communication easier. People can use software like Skype and Hangout to connect with people over Internet. Because of videoconferencing feature, people can connect with audio and video. Computers help families connect to relatives, businesses to organize meetings, companies to take interviews at different locations. Older communications methods like emails are also used.

Human being Vs Computer System

Human being :-

1. Human have feelings, emotions.
2. Have mind , extra intelligence.
3. Can do work by themselves without any command.
4. Can understand the correct and wrong things.
5. Can learn and forget things.
6. Humans are living organisms

Computer :-

1. It has no feeling.
2. It doesn't have intelligence.
3. They follows command.
4. Not able to understand correct and wrong things.
5. Sharp memory store everything.
6. Computers are electronic devices

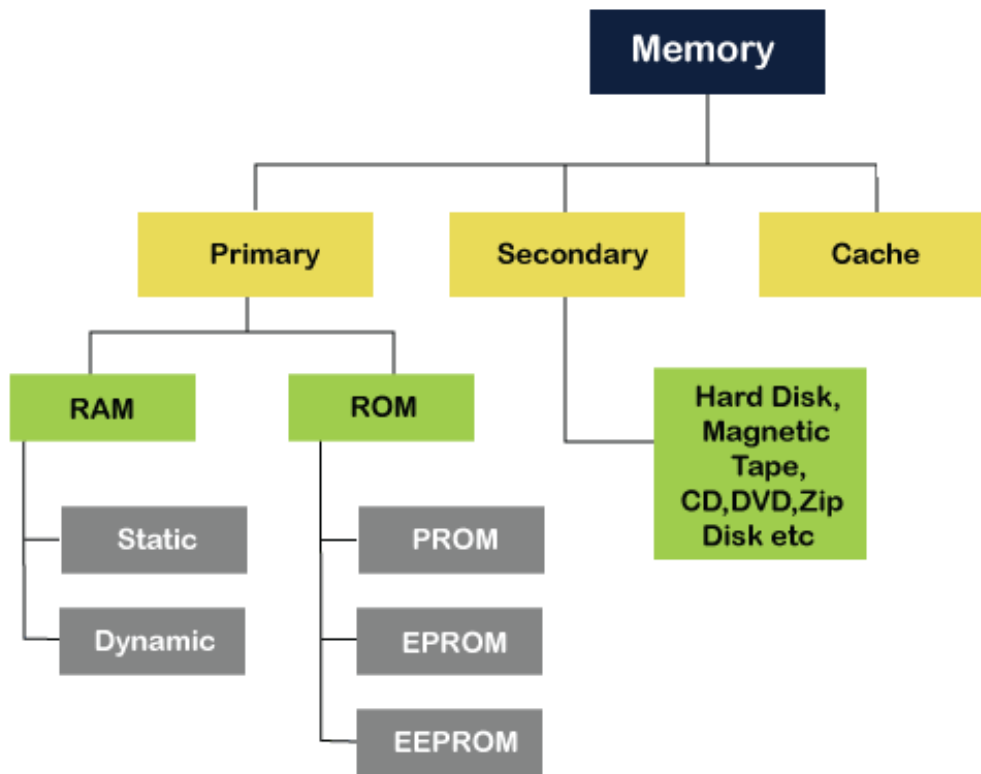
Computer Memory

Computer memory is the storage area used to store data, information or instruction temporarily or permanently. It stores all the inputs before processing and the output are stored after processing of data.

Types of Various Units of Memory

- Bit
- Nibble
- Byte

- Kilo Byte
- MegaByte
- Giga Byte
- Tera Byte
- Peta Byte
- Exa Byte



Primary or Main Memory

- Primary memory is also known as the computer system's main memory that communicates directly within the CPU, and the Cache memory.

- When a program or data is activated to execute, the processor first loads instructions or programs from secondary memory into main memory, and then the processor starts execution.
- Accessing or executing of data from primary memory is faster because it has a cache or register memory that provides faster response, and it is located closer to the CPU.
- The primary memory is volatile, which means the data in memory can be lost if it is not saved when a power failure occurs.
- It is costlier than secondary memory, and the main memory capacity is limited as compared to secondary memory.

The primary memory is further divided into two parts:

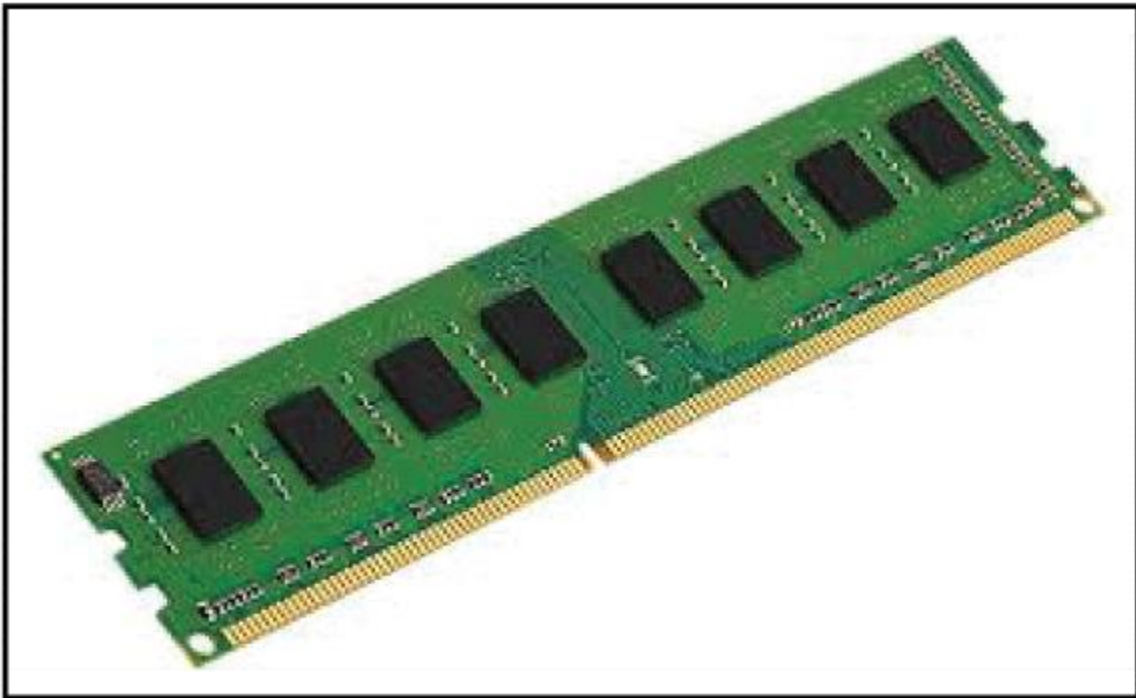
1. RAM (Random Access Memory)

2. ROM (Read Only Memory)

Random Access Memory (RAM)

- RAM is one of the faster types of main memory accessed directly by the CPU.
- It is the hardware in a computer device to temporarily store data, programs or program results.
- It is used to read/write data in memory until the machine is working.

- It is volatile, which means if a power failure occurs or the computer is turned off, the information stored in RAM will be lost.
- All data stored in computer memory can be read or accessed randomly at any time.



There are two types of RAM:

- SRAM
- DRAM

DRAM: DRAM (**D**ynamic **R**andom-**A**ccess **M**emory) is a type of RAM that is used for the dynamic storage of data in RAM. In DRAM, each cell carries one-bit information. The cell is made up of two parts: a **capacitor** and a **transistor**. The size of the capacitor and the transistor is so small, requiring millions of them to store on a single

chip. Hence, a DRAM chip can hold more data than an SRAM chip of the same size. However, the capacitor needs to be continuously refreshed to retain information because DRAM is volatile. If the power is switched off, the data store in memory is lost.

Characteristics of DRAM

1. It requires continuously refreshed to retain the data.
2. It is slower than SRAM
3. It holds a large amount of data
4. It is the combination of capacitor and transistor
5. It is less expensive as compared to SRAM
6. Less power consumption

SRAM: SRMA (**Static Random-Access Memory**) is a type of RAM used to store static data in the memory. It means to store data in SRAM remains active as long as the computer system has a power supply. However, data is lost in SRAM when power failures have occurred.

Characteristics of Static Ram

1. It does not require to refresh.
2. It is faster than DRAM
3. It is expensive.

4. High power consumption
5. Longer life
6. Large size
7. Uses as a cache memory

SRAM	DRAM
It is a Static Random-Access Memory.	It is a Dynamic Random Access Memory.
The access time of SRAM is slow.	The access time of DRAM is high.
It uses flip-flops to store each bit of information.	It uses a capacitor to store each bit of information.
It does not require periodic refreshing to preserve the information.	It requires periodically refreshing to preserve the information.
It uses in cache memory.	It is used in the main memory.
The cost of SRAM is expensive.	The cost of DRAM is less expensive.
It has a complex structure.	Its structure is simple.
It requires low power consumption.	It requires more power consumption.

Read-Only Memory (ROM)

ROM is a memory device or storage medium that is used to **permanently** store information inside a chip. It is a read-only memory that can only read stored information, data or programs, but we cannot write or modify anything. A ROM contains some important instructions

or program data that are required to start or boot a computer. It is a **non-volatile** memory; it means that the stored information cannot be lost even when the power is turned off or the system is shut down.



Types Of ROM

1. **MROM (Masked Read Only Memory):**
MROM is the oldest type of read-only memory whose program or data is pre-configured by the integrated circuit manufacture at the time of manufacturing. Therefore, a program or instruction stored within the MROM chip cannot be changed by the user.

2.PROM (Programmable Read Only Memory):

It is a type of digital read-only memory, in which the user can write any type of information or program only once. It means it is the empty PROM chip in which the user can write the desired content or program only once using the special PROM programmer or PROM burner device; after that, the data or instruction cannot be changed or erased.

3.EPROM (Erasable and Programmable Read Only Memory):

It is the type of read only memory in which stored data can be erased and re-programmed only once in the EPROM memory. It is a non-volatile memory chip that holds data when there is no power supply and can also store data for a minimum of 10 to 20 years. In EPROM, if we want to erase any stored data and re-programmed it, first, we need to pass the ultraviolet light for 40 minutes to erase the data; after that, the data is re-created in EPROM.

4.EEPROM (Electrically Erasable and Programmable Read Only Memory):

The EEROM is an electrically erasable and programmable read only memory used to erase stored data using a high voltage electrical charge and re-programmed it. It is also a non-volatile memory

whose data cannot be erased or lost; even the power is turned off. In EEPROM, the stored data can be erased and reprogrammed up to 10 thousand times, and the data erase one byte at a time.

RAM	ROM
It is a Random-Access Memory.	It is a Read Only Memory.
Read and write operations can be performed.	Only Read operation can be performed.
Data can be lost in volatile memory when the power supply is turned off.	Data cannot be lost in non-volatile memory when the power supply is turned off.
It is a faster and expensive memory.	It is a slower and less expensive memory.
Storage data requires to be refreshed in RAM.	Storage data does not need to be refreshed in ROM.
The size of the chip is bigger than the ROM chip to store the data.	The size of the chip is smaller than the RAM chip to store the same amount of data.
Types of RAM: DRAM and SRAM	Types of ROM: MROM, PROM, EPROM, EEPROM

Secondary Memory

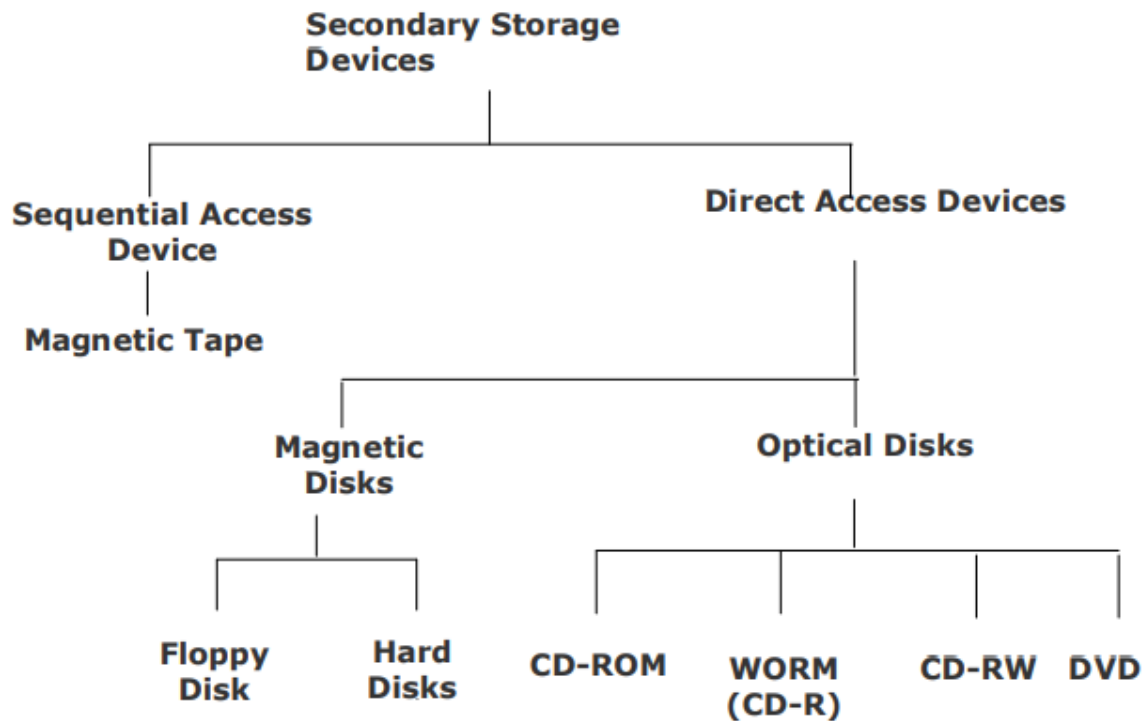
- Secondary memory is a **permanent storage** space to hold a large amount of data.
- Secondary memory is also known as external memory that representing the various storage media (hard drives, USB, CDs, flash drives and DVDs) on

which the computer data and program can be saved on a long term basis.

- It is cheaper and slower than the main memory.
- Unlike primary memory, secondary memory cannot be accessed directly by the CPU.
- Instead of that, secondary memory data is first loaded into the RAM (Random Access Memory) and then sent to the processor to read and update the data.
- Secondary memory devices also include magnetic disks like hard disk and floppy disks, an optical disk such as CDs and CDRoms, and magnetic tapes.

Features of Secondary Memory

- Its speed is slower than the primary/ main memory.
- Store data cannot be lost due to non-volatile nature.
- It can store large collections of different types, such as audio, video, pictures, text, software, etc.
- All the stored data in a secondary memory cannot be lost because it is a permanent storage area; even the power is turned off.
- It has various optical and magnetic memories to store data.



Secondary Storage Devices

Used in a computer system to overcome the limitations of primary storage

Has virtually unlimited capacity because the cost per bit of storage is very low

Used to store large volumes of data on a permanent basis

Also known as auxiliary memory

Sequential Access Storage Device

- 1.It is a class of data storage devices that read stored data in a sequence.
- 2.Arrival at the desired storage location may be preceded by sequencing through other locations
- 3.Data can only be retrieved in the same sequence in which it is stored Access time varies according to the storage location of the information being accessed
- 4.Suitable for sequential processing applications where most, if not all, of the data records need to be processed one after another
- 5.Magnetic tape is a typical example of such a storage

Magnetic Tape

- 1.Plastic ribbon is coated with a magnetizable recording material such as iron-oxide or chromium dioxide
- 2.Data are recorded on the tape in the form of tiny invisible magnetized and non-magnetized spots (representing 1s and 0s) on its coated surface
- 3.Devices that record and play audio and video using magnetic tape are tape recorders and videotape recorders.
- 4.A device that stores data on magnetic tape is known as a ***tape drive***.

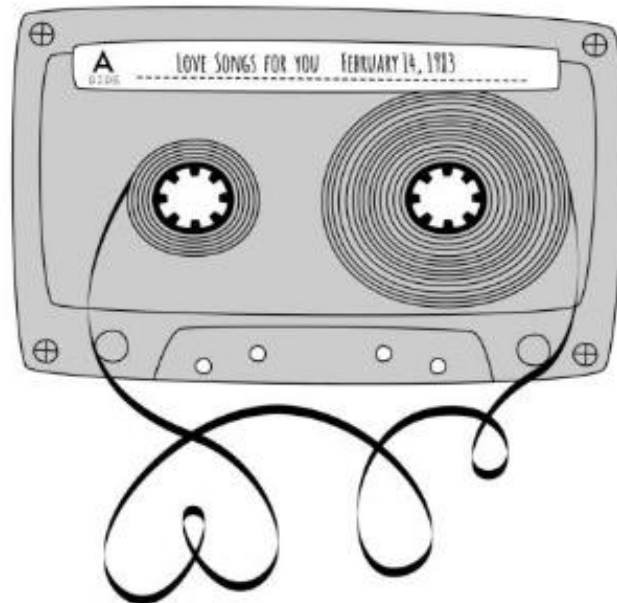
5.A magnetic tape reel/cartridge/cassette has to be first loaded on a tape drive for reading/writing of data on it.

Advantages

- 1.Storage capacity is virtually unlimited because as many tapes as required can be used for storing very large data sets
- 2.Cost per bit of storage is very low for magnetic tapes.
- 3.Tapes can be erased and reused many times
- 4.Tape reels and cartridges are compact and light in weight
- 5.Easy to handle and store.
- 6.Very large amount of data can be stored in a small storage space
- 7.Compact size and light weight
- 8.Magnetic tape reels and cartridges are also easily portable from one place to another
- 9.Often used for transferring data and programs from one computer to another that are not linked together

Disadvantages

1. Due to their sequential access nature, they are not suitable for storage of those data that frequently require to be accessed randomly
2. Must be stored in a dust-free environment because specks of dust can cause tape-reading errors
3. Must be stored in an environment with properly controlled temperature and humidity levels
4. Tape ribbon may get twisted due to warping, resulting in loss of stored data
5. Should be properly labeled so that some useful data stored on a particular tape is not erased by mistake



Direct Access Storage Devices

A direct-access storage device (DASD) is another name for secondary storage devices that store data in discrete locations with a unique address, such as hard disk drives, optical drives and most magnetic storage devices. A direct access storage device is a type of device that supports direct access to stored data.

Magnetic disks:

- i. A magnetic disk is a storage device that uses a magnetization process to write, rewrite and access data.
- ii. It is covered with a magnetic coating.
- iii. Data can be modified or can be deleted easily in the magnetic disk memory.
- iv. It also allows random access to data.
- v. Hard disks and floppy disks are common examples of magnetic disks.

a. Floppy Disk: A floppy disk is a flexible disk with a magnetic coating on it, and it is packaged inside a protective plastic envelope. These are among the oldest portable storage devices that could store up to 1.44 MB of data, but now they are not used due to very little memory storage.



b. Hard Disk Drive (HDD):

- i. Hard disk drive comprises a series of circular disks called ***platters*** arranged one over the other almost $\frac{1}{2}$ inches apart around a ***spindle***.
- ii. stores and retrieves data using magnetic storage with one or more rigid rapidly rotating platters coated with magnetic material.
- iii. The platters are paired with magnetic heads, usually arranged on a moving actuator arm, which read and write data to the platter surfaces.
- iv. Data is accessed in a random-access manner, meaning that individual blocks of data can be stored and retrieved in any order.
- v. HDDs are a type of non-volatile storage

vi.A magnetic reader arm is used to read data from and write data to the disks.

vii.A typical modern HDD has a capacity in terabytes (TB)



Advantages:-

- 1.These are economical memory
- 2.Easy and direct access to data is possible.
- 3.It can store large amounts of data.
- 4.It has a better data transfer rate than magnetic tapes.
- 5.It has less prone to corruption of data as compared to tapes.

Disadvantages:-

- 1.These are less expensive than RAM but more expensive than magnetic tape memories.
- 2.It needs a clean and dust-free environment to store.
- 3.These are not suitable for sequential access.

Optical Disk

An optical disk is an electronic data storage medium that can be written to and read from using a low-powered laser beam. Most of today's optical disks are available: compact disks (CDs), digital versatile disks (DVDs).

- Optical disks rely on a red or blue laser to record and read data. Most of today's optical disks are flat, circular.
- Data is written to an optical disk in a radial pattern starting near the center.
- An optical disk drive uses a laser beam to read the data from the disk.
- An optical disc can store more data and has a longer lifespan
- All modern formats use the same basic sandwich of materials structure.
- The base is formed by using a hard-plastic substrate, and a reflective layer of metallic foil is used for encoding the digital data.
- Next, a layer of clear polycarbonate protects the foil and allows the laser beam to pass through to the reflective layer.

Uses

- Used to transfer data to various devices.
- They are used to deliver the software to others.
- They help users to hold large amounts of data, like videos, photos, music, and more.
- Also, optical media are used to get back up from a local machine.

CD-Compact Discs

Computers can read and write to CDs and DVDs using a CD writer or DVD writer drive.

Drives such as CD-R (compact disc recordable) and DVD-R (digital versatile disc recordable) drives that read and write information to discs are known as MO (magneto-optic).

MO drives, such as CD-R and DVD-R drives, are used to read and write information to discs (magneto-optic)

DVD- Digital Versatile Discs

DVD stand for Digital Versatile Disk and are evolution from CD drives, have significantly more performance and

capacity than CD while maintaining same physical dimensions.

Advantages of Optical Disk

Cost- Only plastics and aluminum foils are used in the production of an optical disk, which makes their manufacturing cost less expensive.

Durability- It is not caused to data losses due to any power failure. Hence, it can run a long time.

Simplicity- With the help of using optical disks, the process of backup of data is much easier.

Portability- Optical disks are very portable. They can be used in different computers and devices and transported to different places as they can be placed inside bags.

Disadvantages of Optical Disk

Security- Optical disks need to keep safe from the hands of thieves. Hence, it may provide insecurity and due to its size, there are more prone to lose and theft.

Reliability- Unlike flash drives, any plastic casings can be caused to damage optical disks. Therefore, they make the disk unreadable as they are prone to scratching the disks. The data stored on an optical disk cannot be recovered anymore.

Capacity- While comparing with any other forms of storage medias, it has less storage capacity. Storage capacity of optical disk is 4.7GB.

Duplication- Like a USB flash drive, it is not easy to make a duplicate copy by using an optical disk. There is needed a separate software and hardware in order to process of burning.

S.NO	CD	DVD
1.	The storage capacity of CD is 700 MB.	While the storage capacity of DVD is 4.7 GB to 17 GB.
2.	In CD, the recording or metal layer is situated on the top of disk.	While in DVD, the recording or metal layer is situated in middle of disk.
3.	There is only single pit layer in CD.	While there are double layers of pits in DVD.

S.NO	CD	DVD
4.	In CD, there is 1.6 micrometer space between the spiral's loops.	While in DVD, there is 0.74 micrometer space between the spiral's loops.
5.	In CD, the data transfer rate is 1.4 MB to 1.6 MB/sec.	While in DVD, the data transfer rate is 11 MB/sec.
6.	It stands for Compact Disk	It stands for Digital Versatile Disk
7.	The thickness of CD is 1.2 mm.	While the thickness of DVD is 0.6 mm.



Features	Primary Memory	Secondary Memory
Definition	The primary memory of a computer is the main memory that is utilized to store data temporarily.	Secondary memory defines to additional storage devices that are utilized to store data permanently.
Accessibility	It is directly accessible by the processor.	It is not directly accessible via the processor.
Nature of Memory	It is both volatile and non-volatile memory.	It is a non-volatile memory in nature.
Other names	It is also known as the main memory of the system.	It is also known as the secondary or auxiliary memory of a computer system.
Formation	It is composed of semiconductors.	It is composed of magnetic and optical materials.
Data	The data that must be executed is copied to the main memory.	It is utilized to store data that requires should be stored permanently.
Access Speed	The speed of accessing data is faster in primary memory.	The speed of accessing data is slower in secondary memory.
Cost	It is more costly than secondary memory.	It is cheaper than primary memory.
Size	The size of primary memory is small.	The size of secondary memory is large.
Memory	It is internal memory.	It is an external memory.
Types	It is mainly of two types: RAM and ROM.	Magnetic memory, semiconductor memory, and optical memory are the three most popular types of secondary memory.
Access	It is accessed via the data bus.	It is accessed via the input-output channel.

Cache Memory

Data in primary memory can be accessed faster than secondary memory but still, access times of primary memory are generally in a few microseconds, whereas the CPU is capable of performing operations in nanoseconds. Due to the time lag between accessing data and acting on data performance of the system decreases as the CPU is not utilized properly, it may remain idle for some time. In order to minimize this time gap new segment of memory is introduced known as Cache Memory.

Role of Cache Memory

- It provides faster access.
- It acts buffer between CPU and main memory(RAM).
- Primary role is to reduce average time taken to access data, improving overall system performance.

Advantages of Cache Memory

1. **Faster access:** Faster than main memory. It resides closer to CPU, typically on same chip.

2. **Reducing memory latency:** Memory access latency refers to time taken for processes to retrieve data from memory. It takes less time to access data.
3. **Lowering bus traffic:** Accessing data from main memory involves transferring it over system bus. Bus is shared resource and excessive traffic can lead to congestion and slower data transfers. By utilizing cache memory, processor can reduce frequency of accessing main memory resulting in less bus traffic and improves system efficiency.
4. **Increasing effective CPU utilization:** Cache memory allows CPU to operate at a higher effective speed. CPU can spend more time executing instruction rather than waiting for memory access. This leads to better utilization of CPU's processing capabilities.
5. **Enhancing system scalability:** Cache memory helps improve system scalability by reducing impact of memory latency on overall system performance.

Disadvantages Of Cache Memory

1. Cache memory is smaller, a large amount of data cannot be stored
2. It is very costly as it is fast memory to access data.

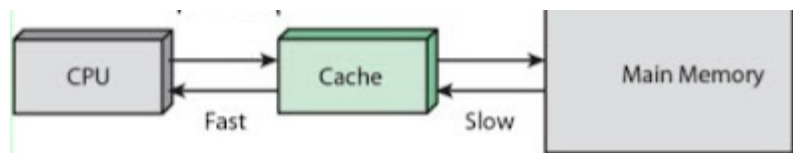
Whenever CPU needs any data it searches for corresponding data in the cache (fast process) if data is found, it processes the data according to instructions, however, if data is not found in the cache CPU search for that data in primary memory (slower process) and loads it into the cache. This ensures frequently accessed data are always found in the cache and hence minimizes the time required to access the data.

Cache Performance

- On searching in the cache if data is found, a cache hit has occurred.
- On searching in the cache if data is not found, a cache miss has occurred.

*Performance of cache is measured by the number of cache hits to the number of searches. This parameter of measuring performance is known as the **Hit Ratio**.*

Hit ratio=(Number of cache hits)/(Number of searches)



Flash Memory

1. Flash memory is secondary memory and so it is not volatile which means it persists the data even if there is not an electrical supply provided.

2. This flash memory works on the principle of EEPROM. EEPROM stands for Electrical Erasable Programmable Read-Only Memory.

3. Flash Memory can be erased multiple times and update the data or program integrated into it.

4. So it gives flexibility to the updation of the program but ROM has no such type of feature.

Features of Flash Memory

- **Non-volatile:** There is no loss of data when there is no electricity supply.
- **Fast access times:** It supports solid-state technology so it has faster access time.
- **Large storage capacity:** Flash memory devices can store large amounts of data, from a few GB (Gigabytes) to several TB (Terabytes).
- **Low power consumption:** It uses less amount of electricity to read the data.

- **Flexibility towards Erase and write operations:** Flash memory can be erased electrically multiple times and read multiple times so flexibility towards read/write operation is more.

Working of Flash Memory

- **Writing the data into flash memory:** Flash memory is made up of small-small memory cells which are made up of floating-gate transistors. All memory cells are organized in the sequential order called an array of memory cells. Data can be stored to the hardware in the form of 0 and 1 only. So to store data into the flash memory we should give electric supply to add the pattern of 0 and 1 into the flash memory and once this pattern is embedded on to the chip then corresponding data is also stored in flash memory. If we want to erase data from flash memory then we have to do it by supplying electrical impulse to make all bits of memory cell to 0.
- **Reading the data into flash memory:** To read the bit which can contain 0 or 1 we have to apply voltage to the gate of transistor, and whatever the current flow in the circuit is measured and by using that measurement we are good to go to identify the bit present in the cell.

Virtual Memory

Virtual Memory is used in the computer memory to increase the storage capacity of main memory. It is a logical storage unit of a computer that creates an illusion to execute a large program that may not be completely placed in main memory. It allows user to load or store data program or files larger than size of main memory.

Advantages of Virtual Memory

1. Virtual Memory allows the users to run more than one application at once.
2. It enhances the degree of multiprogramming.
3. Virtual Memory is a logical unit of computer memory that increases main memory capacity by storing or executing a large size program than main memory.
4. It increases the CPU utilization in virtual memory.
5. It is required whenever the system does not have much space to store any big programs or files.

Disadvantages of Virtual Memory

1. It can slow the process of application in the system.
2. It may take more time to switch between applications.
3. It reduces the stability of the system.

S. N.	Parameter Difference	Cache Memory	Virtual Memory
1.	Definition	Cache Memory is the high speed of computer memory that reduces the access time of files or documents from the main memory.	Virtual Memory is a logical unit of computer memory that increases the capacity of main memory by storing or executing programs of larger size than the main memory in the computer system.
2.	Memory Unit	Cache Memory is defined as a memory unit in a computer system.	Virtual Memory is not defined as a memory unit.
3.	Size	Its size is very small as compared to Virtual Memory.	Its size is very large as compared to the Cache Memory.
4.	Speed	It is a high-speed memory as compared to Virtual Memory.	It is not a high-speed memory as compared to the Cache Memory.
5.	Operation	Generally, it stores frequently used data in the cache memory to reduce the access time of files.	The virtual memory keeps those data or programs that may not completely be placed in the main memory.
6.	Management	Cache Memory is controlled by the hardware of a system.	Whereas the virtual memory is control by the Operating System (OS).
7.	Mapping	It does not require a mapping structure to access the files in Cache Memory.	It requires a mapping structure to map the virtual address with a physical address.